



IMAGINE BUILDING DIFFERENTLY

## PRODUCT DATA SHEET

# 80-Roof Assembly B.PUBLIC Prefab

## PRODUCT OVERVIEW

### Product Description

B.PUBLIC Prefab panelized assemblies exemplify best practices in construction and building science for unparalleled performance, durability, health, and environmental sustainability. The panels are manufactured in controlled environments leveraging both traditional craftsmanship and the latest technology to deliver airtight, vapor-open, and carbon-storing assemblies that achieve Net Zero and Passive House standards. The B.PUBLIC envelope provides industry leading comfort, quiet, and speed to new construction.

### Application

The 80-Roof is designed to meet typical type V construction requirements up to 3 stories coast to coast. Best suited for single-family residences in colder IECC climate regions 5, 6, and 7. B.PUBLIC assemblies and component connections minimize thermal bridging allowing for unprecedented design freedom without sacrificing performance. The B.PUBLIC 80-Roof assembly is available in **Flat** or **Low-slope** configurations.



## TECHNICAL OVERVIEW

|                   |                                                                                           |
|-------------------|-------------------------------------------------------------------------------------------|
| Description       | Offsite prefabricated light wood assembly for flat or low-slope roofs                     |
| R-Value           | 21 1/2" Cellulose / R-3.7 per inch                                                        |
| Vapor Control     | Class II / vapor variable membrane 0.13 to >13.0 perms<br>Rated WRB- 2 months exposure    |
| Cavity Insulation | Dense-pack Cellulose / Borate treated for fire / mold / insect resistance.                |
| Weather Barrier   | Fully adhered monolithic non-porous vapor-open weather barrier / rated 4 months exposure. |
| Thickness         | <b>22 1/8"</b> (23 5/8" plus 11/2" furring for interior service cavity)                   |

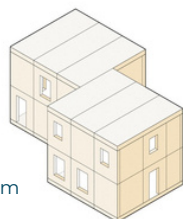


### Installation

All prefabricated assemblies come fully wrapped in rated weather resistant barriers for durability during delivery and installation. Our unique isolated cavity design coupled with a redundant air-barrier ensures that prefabricated assemblies exceed Passive House airtightness requirements.

### Key Attributes

- Breathable yet water-proof vapor-open assembly
- Class A flame and smoke resistant materials
- Sealed to eliminate energy loss due to wind washing
- Grade 1 highest quality installation rating
- Industry leading acoustic performance
- Minimized thermal bridging per PHI standards
- Biogenic materials / Carbon sequestering / Low embodied energy
- > 70% post consumer recycled content
- Living Building Challenge compliant / Red List free
- Urea-Formaldehyde & foam free / No dangerous off-gassing





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## MATERIAL SPECIFICATIONS

### STANDARD ASSEMBLY SPECIFICATIONS\*

|                                   |                                                                                                                                                                                                                                                                                                 |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>                | Offsite prefabricated light wood assembly for flat or low-slope roofs                                                                                                                                                                                                                           |
| <b>R-Value</b>                    | 21 1/2" Cellulose / R-3.7 per inch                                                                                                                                                                                                                                                              |
| <b>Thickness</b>                  | <b>22 1/8"</b> (23 5/8" plus 1 1/2" furring for interior service cavity)                                                                                                                                                                                                                        |
| <b>1 Interior Membrane</b>        | IBC / IRC Class II vapor retarder<br>Vapor variable membrane<br>0.13 to >13.0 perms<br>WRB -rated for 2 months exposure<br>Class A Flame spread index 0<br>Class A Smoke developed index 35<br>ICC -ES report ESR-4854<br>Air permeance 0.00004 cfm/ft                                          |
| <b>2 Cavity Insulation</b>        | Dense-pack Cellulose<br>>3.5 lb per cubic ft.<br>Borate treated for fire / mold / insect resistance.<br>R-3.70 per inch<br>85% recycled content<br>Formaldehyde free.<br>-35 GWP value kg CO2e<br>Class A fire rated                                                                            |
| <b>3 Structural Framing</b>       | 16" APA web joist @ 24" OC with secondary 2x6 framing layer.<br>1 1/8" engineered rim joist<br>Continuous blocking                                                                                                                                                                              |
| <b>4 Structural Sheathing</b>     | 5/8" APA rated sheathing<br>Glued & nailed                                                                                                                                                                                                                                                      |
| <b>5 Exterior Weather Barrier</b> | Fully adhered monolithic non-porous vapor-permeable weather resistant barrier<br>Rated 6 months exposure.<br>Self-sealing around nail & screw penetrations.<br>Waterproof to >32ft water column<br>Vapor permanence 11 perms<br>Airtightness 0.00009 cfm/ft2<br>Class A fire rated FS:10 SDI:15 |
| <b>Dimensional Tolerances</b>     | 3/16" +/- Width & Length<br><1/4" out-of-plane<br><0.3% x length / in-square                                                                                                                                                                                                                    |

